THE MASTERMIND COACHING CLASSES

Revision Notes

Class 10 · Social Science

Geography (Ch 6-10) · CBSE / RBSE Syllabus

HOW TO USE THESE NOTES

Each chapter follows the textbook's own order of sections, so you can read straight through. Bold words are keywords you must be able to define. Tables compare and classify; shaded boxes hold definitions, formulas and key facts. The navy Key Takeaways box at the end of every chapter is your last-two-minutes revision.

CH.	CHAPTER TITLE	CORE THEME
6	Resources and Development	Types of resources, resource planning, land use and soils of India
7	Forest and Wildlife Resources	Biodiversity, endangered species, conservation laws and community efforts
8	Water Resources	Water scarcity, multipurpose projects and rainwater harvesting
9	Agriculture	Farming systems, cropping seasons, major crops and agricultural reforms
10	Minerals and Energy Resources	Minerals, their distribution, conventional and non-conventional energy

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CHAPTER 06 · GEOGRAPHY · CH 1

Resources and Development

Everything in our environment that can be used to satisfy needs, provided it is technologically accessible, economically feasible and culturally acceptable, is a resource. This chapter classifies resources, explains why careful planning and sustainable use are essential, and describes India's land use patterns, soils and soil conservation.

Big Questions

- How are resources classified?
- Why is resource planning needed?
- How is land used in India?
- What are the major soil types and where are they found?
- How can soil be conserved?

Types of Resources

Basis	Types	Examples
Origin	Biotic and abiotic	Forests, animals, fisheries; rocks, metals
Exhaustibility	Renewable and non-renewable	Solar and wind energy, water, forests; minerals and fossil fuels
Ownership	Individual, community, national, international	Plots and houses; village commons, parks; all minerals, water and land within political boundaries; ocean resources beyond 200 nautical miles of the Exclusive Economic Zone
Status of development	Potential, developed, stock, reserves	Wind and solar in Rajasthan and Gujarat (potential); resources already surveyed and in use (developed); water as hydrogen and oxygen (stock); river water used with dams (reserves)

DEFINITION

Sustainable development means development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

Development of Resources

- Resources were thought to be free gifts of nature, and humans used them indiscriminately. This led to depletion of resources, accumulation of resources in a few hands (dividing society into rich and poor) and global ecological crises such as global warming, ozone layer depletion, environmental pollution and land degradation.
- The **Brundtland Commission Report (1987)**, *Our Common Future*, introduced the concept of sustainable development and advocated it as a way ahead.
- The **Rio de Janeiro Earth Summit (June 1992)** in Brazil, with over 100 heads of state, adopted **Agenda 21**, a declaration to achieve sustainable development in the twenty-first century by combating environmental damage, poverty and disease through global cooperation.
- Gandhiji's words: ‘There is enough for everybody's need and not for anybody's greed.’

Resource Planning

Planning is essential for judicious use, especially in a country like India with diverse resources but also regional imbalance. For example, Jharkhand, Chhattisgarh and Madhya Pradesh are rich in minerals and coal; Arunachal Pradesh has abundant water but lacks infrastructure; Rajasthan is rich in solar and wind energy but short of water; Ladakh is rich in cultural heritage but lacks water, infrastructure and minerals.

Step 1
Identify and inventory resources across regions through surveying, mapping and qualitative and quantitative estimation.

Step 2
Evolve a planning structure with appropriate technology, skill and institutional set-up.

Step 3
Match the resource development plans with overall national development plans.

IMPORTANT

Conservation is not hoarding. It means using resources wisely, avoiding wastage and protecting them for the future.

Land Resources and Land Use

India has a land area of about **3.28 million sq km**, and varied relief: mountains (about 30 per cent), plains (43 per cent) and plateaus (27 per cent). Land is used for forests, land not available for cultivation, other uncultivated land (excluding fallow) and fallow lands, and net sown area.

Category	Description
Forests	Area under tree cover; forest area is not the same as forest cover
Land not available for cultivation	Barren and wasteland; land put to non-agricultural uses (buildings, roads, factories)
Other uncultivated land	Permanent pastures, grazing lands, cultivable wasteland (left uncultivated for more than five years), land under tree crops
Fallow lands	Current fallow (left without cultivation for one or less than one agricultural year) and other than current fallow (one to five years)
Net sown area	Area sown with crops in a year; gross cropped area = net sown area plus area sown more than once

- Land use is determined by physical factors (topography, climate, soil types) and human factors (population density, technological capability, culture and traditions).
- **Land degradation:** Mining in states such as Jharkhand, Chhattisgarh, Madhya Pradesh and Odisha leaves deep scars. Overgrazing in Gujarat and Rajasthan, over-irrigation in Punjab, Haryana and western Uttar Pradesh (waterlogging, salinity and alkalinity), and mineral processing (dust) cause degradation.
- **Remedies:** afforestation, proper management of grazing, planting shelter belts, stabilisation of sand dunes with thorny bushes, control of mining activities and proper discharge of industrial wastes.

Soil as a Resource

Soil is the most important renewable resource. It is the medium of plant growth and supports different living organisms. Its formation depends on relief, parent rock, climate, vegetation and time; soil formation is a very

slow process.

Soil	Where	Features and crops
Alluvial	Northern plains from Punjab to Assam, Gujarat via Rajasthan corridor, east coast deltas; deposited by the Indus, Ganga, Brahmaputra	Most widespread and fertile; rich in potash, phosphoric acid and lime; ideal for sugarcane, paddy, wheat and other crops; *Bhangar* (old, kankar nodules) and *Khadar* (new, more fertile)
Black (Regur)	Deccan Trap: Maharashtra, Madhya Pradesh, Gujarat, Andhra Pradesh, Telangana, Karnataka	Formed from lava; clayey, retains moisture; self-ploughing; develops cracks in dry season; rich in calcium carbonate, magnesium, potash and lime but poor in phosphorus; ideal for cotton (black cotton soil)
Red and yellow	Eastern and southern Deccan Plateau; Odisha, Chhattisgarh, middle Ganga plain, Western Ghats piedmont	Develops on crystalline igneous and metamorphic rocks in low-rainfall areas; red due to diffusion of iron, yellow when hydrated; generally poor in nitrogen and humus
Laterite	High temperature and heavy rain areas: Karnataka, Kerala, Tamil Nadu, Madhya Pradesh, Assam hills	Leaching; humus low; suitable for tea, coffee, cashew with manures; used for making bricks
Arid	Rajasthan and dry areas	Sandy, saline; low humus; moisture is low; *kankar* layer limits infiltration; with irrigation can be cultivated
Forest	Hilly and mountainous areas	Loamy and silty in valleys; coarse in upper slopes; acidic with low humus in snow-covered areas

Soil Erosion and Conservation

- **Causes:** deforestation, overgrazing, construction and mining, running water, wind, glaciers.
- **Gully erosion:** running water cuts deep channels; **badlands** are formed (e.g. the Chambal ravines). **Sheet erosion:** water flows as a sheet over a large area and washes away the topsoil.
- **Measures: contour ploughing** (ploughing along contour lines), **terrace farming** (steps on slopes, as in the Himalayas and Western Ghats), **strip cropping** (strips of grass left between crops), and **shelter belts** (rows of trees planted to check wind, in coastal and dry areas).

Quick Revision: Likely Exam Questions

Likely question	Model answer
Define a resource.	Anything in the environment that is technologically accessible, economically feasible and culturally acceptable to meet needs.
What is sustainable development?	Meeting present needs without harming the needs of future generations.
What was Agenda 21?	Declaration of the 1992 Rio Earth Summit for sustainable development in the 21st century.
Which soil is best for cotton and why?	Black soil, because it is moisture-retentive and rich in lime and potash.
Differentiate Khadar and Bhangar.	Khadar is newer, finer and more fertile; Bhangar is older with kankar nodules.
What is contour ploughing?	Ploughing along contour lines to slow runoff and erosion.

Likely question	Model answer
Name two causes of land degradation.	Mining and overgrazing; over-irrigation causes salinity.

★ KEY TAKEAWAYS

- Resources are classified by origin, exhaustibility, ownership and development status.
- Sustainable development was highlighted by the **Brundtland Report (1987)** and **Agenda 21 (Rio, 1992)**.
- Resource planning needs inventory, structure and matching with national plans.
- India's main soils are **alluvial, black, red and yellow, laterite, arid and forest**.
- Erosion is checked by contour ploughing, terraces, strip cropping and shelter belts.

CHAPTER 07 · GEOGRAPHY · CH 2

Forest and Wildlife Resources

India is one of the world's richest countries in biodiversity, yet many of its plants and animals are under threat. This chapter explains the categories of species, the causes of loss, the steps taken to conserve forests and wildlife, and the role played by communities.

Big Questions

- Why is India rich in biodiversity?
- How are species classified by the IUCN?
- What are the causes of depletion?
- How are forests and wildlife conserved?
- How do communities help?

Biodiversity in India

- India has nearly **8 per cent** of the world's total number of species, estimated at 1.6 million.
- India has over 47,000 plant species and about 89,000 animal species, and is also home to a large number of crops and their wild relatives, domesticated animals and breeds.
- Many species are under threat: about 10 per cent of known wild flora and 20 per cent of mammals are on the threatened list; some, such as the **cheetah**, **pink-headed duck** and **mountain quail**, have gone extinct from India. The cheetah was reintroduced in 2022 at Kuno National Park in Madhya Pradesh.

Classification of Species (IUCN)

Category	Meaning	Examples
Normal species	Population levels are normal for survival	Cattle, sal, pine, rodents
Endangered species	In danger of extinction; survival is difficult if the factors causing decline continue	Black buck, crocodile, Indian wild ass, Indian rhino, lion-tailed macaque, sangai (brow-antlered deer)
Vulnerable species	Population declining; likely to move into the endangered category	Blue sheep, Asiatic elephant, Gangetic dolphin
Rare species	Small population, may move into endangered or vulnerable	Himalayan brown bear, wild Asiatic buffalo, desert fox, hornbill
Endemic species	Found only in some areas, usually isolated by natural barriers	Andaman teal, Nicobar pigeon, Andaman wild pig, mithun in Arunachal Pradesh
Extinct species	No longer found anywhere	Asiatic cheetah, pink-headed duck

The IUCN is the International Union for Conservation of Nature and Natural Resources; its Red Data Book keeps a record of endangered species.

Why Are Forest and Wildlife Resources Declining?

- **Colonial expansion:** railways, agriculture, commercial and scientific forestry and mining; large areas of forest were cleared after 1951 for river valley projects, agriculture and plantations. Between 1951 and 1980, over 26,200 sq km of forest area was converted into agricultural land.
- **Shifting cultivation** ('slash and burn') in the north-east and parts of central India.
- **Large development projects**: the Narmada Sagar and others submerge forests; **mining**, such as in West Bengal's Buxa Tiger Reserve, threatens habitats.
- **Grazing and fuelwood collection**, **poaching and illegal trade** in skins, bones and horns, and **forest fires**.
- **Environmental destruction is not just a result of the poor** but of the lifestyles of the rich who consume more; the unequal share of resources leads to depletion.

IMPORTANT

Habitat destruction, hunting, over-exploitation, environmental pollution, poisoning and forest fires are the main factors in biodiversity loss. Loss of species also reduces the health of the ecosystem and affects human well-being, especially of the forest-dwelling poor and women.

Conservation of Forest and Wildlife

- Conservation preserves the ecological diversity (water, air, soil) and our life support systems, and genetic diversity of plants and animals for better growth of species and breeds.
- The **Indian Wildlife (Protection) Act of 1972** provided for an all India list of protected species, banned hunting, protected habitats and curbed trade; many national parks and wildlife sanctuaries were set up.
- **Project Tiger (1973)** was launched to protect the tiger; the tiger population had dropped sharply from 55,000 in the early twentieth century to 1,827 in 1973. Tiger reserves include Corbett (Uttarakhand), Sunderbans (West Bengal), Bandhavgarh (Madhya Pradesh), Sariska (Rajasthan), Manas (Assam) and Periyar (Kerala).
- Other species have schemes of protection, such as the one-horned rhino in Assam and Bengal, and the Asiatic lion in the Gir forest of Gujarat.

▸ Forest categories (as per the Indian Forest Policy)

Type	Features
Reserved forests	Most valuable for conservation; more than half of the total forest land; no grazing or hunting without permission
Protected forests	Nearly one-third; protected from further depletion; livelihood use is allowed
Unclassed forests	Other forests and wastelands belonging to government and private individuals and communities

Reserved and protected forests are also called **permanent forest estates**, maintained for timber and other forest produce. **Madhya Pradesh** has the largest area under permanent forests.

Community and Conservation

- In many parts of India, local communities protect forests because their livelihoods depend on them: villages in Sariska's **Alwar district (Rajasthan)** fought against mining and declared about 1,200 hectares of forest as **Bhairodev Dakav 'Sonchuri'** and made their own rules to protect wildlife.

- **Sacred groves** are patches of forest preserved by tradition with beliefs of deities; trees and animals in them are protected.
- The **Chipko movement** in the Himalayas (Tehri-Garhwal) resisted deforestation; the **Beej Bachao Andolan** in Tehri and **Navdanya** showed that diverse crops can be grown without chemicals.
- **Joint Forest Management (JFM)** began in **1988** when Odisha passed the first resolution; local communities work with the forest department to manage and restore degraded forests and get a share of the produce.
- Religious beliefs protect species such as the **peepal and tulsi**, **black buck** among the Bishnoi people of Rajasthan, and the **Himalayan yew** (Taxus wallichiana), a medicinal plant, found in Himachal and Arunachal Pradesh, which yields taxol for cancer treatment and is endangered by over-harvesting.

Quick Revision: Likely Exam Questions

Likely question	Model answer
What is an endemic species?	A species confined to a particular area, e.g. the Andaman teal.
What is the IUCN?	International Union for Conservation of Nature, which classifies species.
State two causes of depletion of forests.	Agricultural expansion and large development projects; also grazing, fires, mining.
When was Project Tiger launched?	1973.
What is JFM?	Joint Forest Management, in which communities and forest departments protect forests; started in Odisha in 1988.
Name the categories of forests.	Reserved, protected and unclassed.
Why is the Himalayan yew important?	It yields taxol, an anti-cancer drug, and is threatened.

★ KEY TAKEAWAYS

- India has about **8 per cent** of world species, but many are threatened.
- The IUCN groups species as **normal, endangered, vulnerable, rare, endemic and extinct**.
- Forests decline due to agricultural expansion, development projects, mining, grazing and poaching.
- The **Wildlife Protection Act 1972** and **Project Tiger 1973** are key measures.
- Communities, **Chipko**, sacred groves and **JFM (1988)** show participatory conservation.

CHAPTER 08 · GEOGRAPHY · CH 3

Water Resources

About three-fourths of the earth is covered with water, yet only a small part is fresh and usable. This chapter explains why water scarcity occurs even in a water-rich country like India, how dams and multipurpose projects try to solve it, and how traditional and modern rainwater harvesting can help.

Big Questions

- Why is there scarcity of water?
- What are the uses and problems of multipurpose river projects?
- What is rainwater harvesting and how is it practised in India?

Water Scarcity and the Need for Conservation

- About **96.5 per cent** of the world's water is in the oceans and only about **2.5 per cent** is freshwater; of this, nearly 70 per cent is in the form of ice sheets and glaciers. India receives about **4 per cent** of the global precipitation yet it ranks only about 133rd in the world in water availability per person per year, and the availability is uneven in space and time.
- **Water scarcity** is often the result of **over-exploitation, excessive use and unequal access** to water among different social groups, not only of low rainfall. It also arises from a large and growing population, where more people mean more water for domestic use and greater demand for food.
- Irrigation: to increase food production, India has used large amounts of water for the **green revolution**; farmers have used tube wells and pumps, which has led to a fall in groundwater levels.
- Industrialisation and urbanisation: industries are heavy users of water and also pollute it; urban centres have large populations and lifestyles which need more water and energy; housing societies have their own groundwater pumping devices.
- **Water quality** also reduces availability: untreated domestic and industrial waste, chemicals and fertilisers pollute water, and water scarcity is also due to pollution.

Multipurpose River Projects and Integrated Water Resources Management

Dams were traditionally built to impound rivers and rainwater for irrigation. Modern dams serve **multiple purposes**: irrigation, electricity generation, water supply, flood control, recreation, inland navigation and fish breeding. Jawaharlal Nehru proclaimed dams as the **'temples of modern India'**, because they would integrate agricultural development with industrial growth and the growth of the village economy with the urban economy.

Project	River	State
Bhakra-Nangal	Satluj	Punjab/Himachal Pradesh
Hirakud	Mahanadi	Odisha
Nagarjuna Sagar	Krishna	Telangana/Andhra Pradesh
Tungabhadra	Tungabhadra	Karnataka/Andhra Pradesh
Sardar Sarovar	Narmada	Gujarat
Tehri	Bhagirathi	Uttarakhand

▸ Problems and protest

- Dams regulate and sometimes **fragment rivers**, affecting aquatic life and sediment flow; reservoirs flood agricultural land, forests and villages, displacing local communities. Sedimentation reduces storage capacity.
- Dams caused **inter-state water disputes** (such as over the Krishna-Godavari and Cauvery) and in some cases flooding rather than flood control (for example the Maharashtra-Gujarat floods of 2006, linked to water released from dams).
- **Social movements:** the **Narmada Bachao Andolan** and the **Tehri Dam Andolan** opposed dams, mainly for displacement and the cost to the environment. Many such dams did not give the benefits promised: irrigation was used for commercial crops, and water was not equitably shared.
- Hence emphasis shifted to **small, decentralised** projects, using watershed management and rainwater harvesting.

Rainwater Harvesting

> DEFINITION
> **Rainwater harvesting** is the collection and storage of rainwater, mostly in structures, so that it can be used later or can recharge groundwater.

In ancient India, people developed ingenious systems to store water. In the hill and mountainous regions, people built diversion channels like **guls** or **kuls** (Western Himalayas) for agriculture. **Rooftop rainwater harvesting** was commonly practised to store drinking water, particularly in Rajasthan. In the flood plains of Bengal, people developed **inundation channels** to irrigate their fields. In arid and semi-arid regions, agricultural fields were converted into rain-fed storage structures that allowed the water to stand and moisten the soil, e.g. **khadins** in Jaisalmer and **johads** in other parts of Rajasthan.

Region	Traditional method
Western Himalayas (Himachal)	Kuls/guls: diversion channels for irrigation
Rajasthan (Bikaner, Phalodi, Barmer)	**Tankas / kunds**: underground tanks, often built inside the house or courtyard, to store rooftop water; sometimes connected with the roof; ‘palar pani’ is considered the purest form of natural water
Jaisalmer	Khadins; Johads elsewhere in Rajasthan
Bengal	Inundation channels
Meghalaya	Bamboo drip irrigation, about 200 years old: bamboo pipes tap springs and carry water to fields, even over hundreds of metres, about 18-20 litres entering the system is reduced to some 20-80 drops a minute at the plant
Bihar	Ahars and pynes

- **Tamil Nadu** is the first state in India to make rooftop rainwater harvesting structures compulsory for all houses; the states of Karnataka also encouraged it. In **Gendathur**, a remote village in Mysuru, Karnataka, villagers installed rooftop rainwater harvesting in many houses; the village is known as a water-rich village.
- Traditional systems gave way to **piped water and tubewells** and were neglected; **revival** of these methods is needed because they are cheap and ecofriendly and recharge groundwater.

- In **Rajasthan**, the tanka was used for generations; in Bikaner, Phalodi and Barmer almost all the houses traditionally had underground tanks. **Indira Gandhi Canal** brought water to the desert, so tankas were neglected.

> **REMEMBER**
> Conserving water means reducing demand, avoiding pollution, recycling, reviving traditional methods and managing watersheds, and not only building dams.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Why does India face water scarcity despite high rainfall?	Uneven availability, over-use, population pressure and pollution.
What are multipurpose projects?	Dams serving irrigation, power, flood control, supply and navigation.
Who called dams the temples of modern India?	Jawaharlal Nehru.
Name two problems caused by dams.	Displacement and loss of land; sedimentation and inter-state disputes.
What is a tanka?	An underground tank in Rajasthan for storing rainwater.
Describe bamboo drip irrigation.	Meghalaya method of using bamboo pipes to lead spring water to plants.
Which state made rooftop harvesting compulsory?	Tamil Nadu.

★ KEY TAKEAWAYS

- Only about **2.5 per cent** of the world's water is fresh; India's water is unevenly available.
- Scarcity comes from **over-use, population growth, irrigation, industry and pollution**.
- Multipurpose projects bring benefits but also **displacement and conflict**.
- **Rainwater harvesting**, such as tankas, johads and kuls, is a cheap, sustainable answer.
- Revival of traditional methods and decentralised management is the way forward.

CHAPTER 09 · GEOGRAPHY · CH 4

Agriculture

India is an agrarian economy: about half of its people depend on agriculture for a living. This chapter describes the types of farming, the three cropping seasons, the main food, fibre, beverage and plantation crops with their conditions and regions, and the technological and institutional changes in Indian farming.

Big Questions

- What are the types of farming?
- What are rabi, kharif and zaid crops?
- Which crops grow where and under what conditions?
- How have reforms changed farming?
- What challenges does agriculture face?

Types of Farming

Type	Features
Primitive subsistence	Small patches of land, primitive tools (hoe, dao, digging stick), family or community labour, depends on monsoon and natural fertility; also called **slash and burn**: trees are cut and burnt, crops grown, and when fertility falls, the plot is abandoned. Local names: *jhumming* (North-east), *milpa* (Mexico), *conuco* (Venezuela), *roca* (Brazil), *masole* (Central Africa), *ladang* (Indonesia), *ray* (Vietnam)
Intensive subsistence	High pressure of population on land; high doses of biochemical inputs and irrigation; right of inheritance leads to division of land into small plots
Commercial	Higher doses of modern inputs (HYV seeds, fertilisers, pesticides); the degree of commercialisation varies: rice is commercial in Haryana and Punjab but subsistence in Odisha
Plantation	Commercial farming over large area with capital and migrant labour; single crop; links to industry. Tea, coffee, rubber, sugarcane and banana in Assam, West Bengal, Karnataka, Kerala and Tamil Nadu

Cropping Pattern

Season	Sowing and harvest	Crops
Rabi	Sown in winter from **October to December**; harvested in summer from **April to June**	Wheat, barley, peas, gram, mustard; winter rain from western temperate cyclones helps; Punjab, Haryana, Himachal, Uttarakhand, Uttar Pradesh, Bihar, Rajasthan
Kharif	Sown with the onset of monsoon in **June-July**, harvested in **September-October**	Paddy, maize, jowar, bajra, tur (arhar), moong, urad, cotton, jute, groundnut, soyabean; Assam, West Bengal, Odisha, coastal Andhra, Tamil Nadu, Kerala, Maharashtra
Zaid	Short season between rabi and kharif, in **summer**	Watermelon, muskmelon, cucumber, vegetables and fodder crops; sugarcane takes almost a year

Major Crops

Crop	Conditions	Major states and notes
Rice	Kharif; high temperature (above 25 degrees C), high humidity, annual rainfall above 100 cm; irrigation in low-rainfall areas	West Bengal, Uttar Pradesh, Punjab, Andhra Pradesh, Odisha; India is second to China
Wheat	Rabi; cool growing season and bright sunshine at ripening; 50-75 cm rainfall	Ganga-Satluj plains, Deccan black soil; Uttar Pradesh, Punjab, Haryana, Madhya Pradesh, Rajasthan
Millets (jowar, bajra, ragi)	Coarse grains rich in nutrients; need little rain; jowar rain-fed in Maharashtra; bajra in sandy and shallow black soil; ragi on red, black, sandy, loamy soils	Jowar: Maharashtra; bajra: Rajasthan; ragi: Karnataka; 2023 was declared the **International Year of Millets**
Maize	Kharif; temperature 21-27 degrees C; used as food and fodder	Karnataka, Madhya Pradesh, Bihar
Pulses	Leguminous, so they restore soil fertility by fixing nitrogen; need less moisture	India is the largest producer and consumer; Madhya Pradesh, Rajasthan, Maharashtra, Uttar Pradesh, Karnataka
Sugarcane	Tropical and subtropical; hot and humid climate with 75-100 cm rainfall; labour-intensive	Uttar Pradesh, Maharashtra, Karnataka, Tamil Nadu; India is second to Brazil
Oilseeds	Groundnut (kharif; half of oilseed area), mustard (rabi), sesamum, castor seed, linseed, soyabean, sunflower	Gujarat, Andhra Pradesh, Tamil Nadu
Tea	Tropical and subtropical; deep, fertile, well-drained soil; warm moist frost-free climate; labour-intensive	Assam, hills of Darjeeling and Jalpaiguri, Tamil Nadu, Kerala
Coffee	Arabica variety brought from Yemen; shade crop	Nilgiri area of Karnataka, Kerala and Tamil Nadu (Baba Budan Hills)
Rubber	Equatorial; over 200 cm rain and temperature above 25 degrees C	Kerala, Tamil Nadu, Karnataka, Andaman and Nicobar, Garo Hills
Cotton	Black soil; high temperature, light rainfall or irrigation, 210 frost-free days, bright sun; kharif	Maharashtra, Gujarat, Madhya Pradesh, Karnataka, Andhra Pradesh, Tamil Nadu, Punjab, Haryana; India is among top producers
Jute	‘Golden fibre’; well-drained fertile soil of flood plains, high temperature during growth	West Bengal, Bihar, Assam, Odisha, Meghalaya; used for gunny bags, mats, rope

India is also a leading producer of **fruits and vegetables**: mangoes in Maharashtra, Andhra Pradesh, Uttar Pradesh and West Bengal; oranges in Nagpur and Cherrapunji; bananas in Kerala, Mizoram, Maharashtra and Tamil Nadu; and peas, cauliflower, onions, potatoes, tomatoes and brinjal. **Horticulture** is the growing of fruits and vegetables.

Technological and Institutional Reforms

- **Land reforms** after independence: collectivisation, consolidation of holdings, cooperation and abolition of zamindari. The land ceiling act and other measures were not fully implemented because of the vested interests of political leaders.
- **Green Revolution (1960s-70s):** HYV seeds, fertilisers and irrigation raised wheat and rice output in Punjab, Haryana, western Uttar Pradesh; the **White Revolution** (Operation Flood) did the same for milk. It also widened regional and income disparities.
- **Bhoodan-Gramdan:** Vinoba Bhave's Bhoodan movement (first gift in Pochampalli in 1951) asked landlords to donate land to the landless.
- **Support measures:** Minimum Support Price, remunerative and procurement prices, crop insurance, the Kisan Credit Card (KCC) and Personal Accident Insurance Scheme (PAIS), special weather bulletins and TV programmes for farmers, and subsidised inputs.

Contribution of Agriculture and Challenges

- Agriculture provides food, raw materials for industry (cotton, jute, sugar) and export earnings (tea, coffee, spices), and employs nearly half of India's workers; its share in GDP has fallen as other sectors have grown.
- **Challenges:** small and fragmented holdings, dependence on monsoon, lack of irrigation, soil degradation, falling groundwater, rising input costs, competition from imports and farmers' debt.
- **Remedies:** better irrigation, organic and sustainable farming, crop diversification, **millets and pulses**, food processing and fair prices.

Quick Revision: Likely Exam Questions

Likely question	Model answer
What is jhumming?	Slash and burn shifting agriculture of the north-east.
Name the three cropping seasons.	Rabi, kharif and zaid.
Which crops are rabi?	Wheat, barley, peas, gram and mustard.
Why are pulses good for soil?	They are leguminous and fix nitrogen.
Which crop is called the golden fibre?	Jute.
What is a plantation?	Large-scale single-crop commercial farming with capital and migrant labour.
What was the Green Revolution?	Use of HYV seeds, fertilisers and irrigation to raise wheat and rice output.

★ KEY TAKEAWAYS

- Farming is **primitive subsistence, intensive subsistence, commercial or plantation**.
- **Rabi** (Oct-Dec sowing), **kharif** (June-July sowing) and **zaid** (summer) are the three seasons.
- Rice, wheat, millets, pulses, sugarcane, oilseeds, tea, coffee, rubber, cotton and jute need different conditions.
- Reforms such as **Green Revolution**, land reforms and **MSP** changed Indian farming.
- Fragmentation, monsoon dependence and falling groundwater remain challenges.

CHAPTER 10 · GEOGRAPHY · CH 5

Minerals and Energy Resources

Minerals are the building blocks of industry and energy resources run it. This chapter explains what minerals are, where India's iron ore, manganese, copper, bauxite, mica and others occur, and how coal, petroleum, gas, electricity and non-conventional sources meet India's energy needs, together with the need to conserve them.

Big Questions

- What are minerals and how do they occur?
- Where are India's important minerals found?
- What are conventional and non-conventional energy sources?
- Why must minerals and energy be conserved?

What Is a Mineral?

A mineral is a homogeneous, naturally occurring substance with a definable internal structure. Minerals are found in varied forms in nature, from the hardest diamond to the softest talc. They generally occur in **ores**, the accumulation of any mineral mixed with other elements.

- **Mode of occurrence:** in **igneous and metamorphic rocks** they are found in cracks, crevices, faults or joints (veins and lodes; e.g. tin, copper, zinc, lead); in **sedimentary rocks** in beds or layers, e.g. coal, iron ore, gypsum, potash salt, sodium salt; by **decomposition of surface rocks** in residual mass, e.g. bauxite; as **alluvial deposits** called placer deposits, e.g. gold, silver, tin, platinum; in **ocean waters**, e.g. common salt, magnesium, bromine; manganese nodules on the ocean floor.
- **Extraction:** mining (underground, deep, called shaft mining), open-cast or quarrying (cheapest), and drilling for petroleum and natural gas.

Ferrous and Non-Ferrous Minerals

Mineral	Features	Distribution in India
Iron ore	Basic mineral, backbone of industrial development; **magnetite** (best, over 70% iron, magnetic) and **haematite** (most important in quantity, about 60% iron)	Odisha-Jharkhand belt (Badampahar, Gua), Durg-Bastar-Chandrapur belt in Chhattisgarh and Maharashtra (Bailadila exports to Japan and South Korea via Visakhapatnam), Ballari-Chitradurga-Chikkamagaluru-Tumakuru belt in Karnataka (Kudremukh), Maharashtra-Goa belt
Manganese	Used in steel-making and ferro-manganese alloys (about 10 kg for a tonne of steel); also bleaching powder, insecticides, paints	Odisha is the leading producer
Copper	Malleable, ductile, a good conductor; used in electrical cables and electronics	Balaghat (Madhya Pradesh), Khetri (Rajasthan), Singhbhum (Jharkhand)
Bauxite	Ore of aluminium; light, strong; clay-like; formed by decomposition of rocks	Amarkantak plateau, Maikal hills, Bilaspur-Katni plateau; **Odisha** is the largest producer, Panchpatmali in Koraput

Mineral	Features	Distribution in India
Mica	Transparent, black, green, red, yellow or brown; excellent insulator, used in electrical and electronic industries	Koderma-Gaya-Hazaribagh belt in Jharkhand, Ajmer in Rajasthan, Nellore in Andhra Pradesh
Limestone	Found in sedimentary rocks of calcium carbonate; basic raw material for cement; also used in smelting iron ore	Widely distributed

NOTE

India has uneven distribution of minerals: about **97 per cent of coal** reserves occur in the Damodar, Sone, Mahanadi and Godavari valleys; **petroleum** reserves are in the Mumbai High, Gujarat and Assam; most metallic minerals lie in the old rocks of the Peninsular plateau, while limestone and other non-metallic minerals are found in sedimentary rocks on the western and eastern flanks.

Conventional Sources of Energy

Source	Notes
Coal	Most abundant fossil fuel in India; used for power generation and industry. Types by carbon content (from least to most): **peat** (low carbon, high moisture), **lignite** (low-grade brown coal, soft, high moisture; Neyveli, Tamil Nadu), **bituminous** (most popular in commercial use; metallurgical grade for blast furnace), **anthracite** (highest quality, hard). Gondwana coal (about 200 million years old) in Damodar valley (Jharia, Raniganj, Bokaro), Godavari, Mahanadi, Sone; Tertiary coal (about 55 million years old) in north-eastern states
Petroleum	Provides fuel for heat and lighting, lubricants and raw material for industries; mostly found in anticlines and fault traps in sedimentary rocks. About 63 per cent of India's production is from **Mumbai High**, 18 per cent from Gujarat and 16 per cent from Assam. **Digboi** (Assam) is the oldest oil-producing area. Ankleshwar is the most important field in Gujarat. Refineries are ‘nodal industries’ for synthetic textiles, fertiliser and chemicals
Natural gas	Found with or without petroleum; used as fuel and as industrial raw material (CNG, PNG); large reserves in Krishna-Godavari basin; Jaisalmer, Mumbai High, Gulf of Khambhat; **HBJ pipeline** links Hazira (Gujarat), Bijaipur (Madhya Pradesh) and Jagdishpur (Uttar Pradesh)
Electricity	Generated by running water turning turbines (**hydro**) and by burning coal, petroleum or natural gas (**thermal**). Hydroelectricity: Bhakra-Nangal, Damodar Valley Corporation, Kopili Hydel project. Thermal power stations use coal, petroleum and natural gas
Nuclear	Uranium and thorium; Jharkhand and Aravalli ranges of Rajasthan; thorium in monazite sands of Kerala. Power stations: Tarapur (Maharashtra), Rawatbhata (Rajasthan), Kalpakkam (Tamil Nadu), Narora (Uttar Pradesh), Kaiga (Karnataka), Kakrapar (Gujarat)

Non-Conventional Sources of Energy

- **Solar energy:** photovoltaic technology converts sunlight directly into electricity; India is a tropical country with ample sunshine. The **Bhuj** (Gujarat) plant is an important solar plant. Solar energy is expected to replace kerosene and firewood in rural areas.
- **Wind energy:** India has potential in Gujarat, Rajasthan and Tamil Nadu; the largest wind farm cluster is from **Nagercoil to Madurai** in Tamil Nadu. India ranks high in the world in installed wind power capacity.

- **Biogas:** organic waste such as cattle dung, agricultural waste and sewage produces gas; ‘gobar gas plants’ provide energy for cooking and lighting in rural areas.
- **Tidal energy:** high and low tides turn turbines; sites are the Gulf of Khambhat, Gulf of Kutch and the Sunderbans delta. **Geothermal energy:** heat from the earth, at Parvati valley near Manikaran (Himachal Pradesh) and the Puga Valley in Ladakh.

Conservation of Mineral and Energy Resources

- Minerals take millions of years to form, but their consumption is rapid: **the rate of replenishment is infinitely small** compared with the present rate of consumption. Mineral resources are finite and non-renewable.
- Measures: use of minerals in a **planned and sustainable manner**; improved technologies to allow use of low-grade ores at low cost; **recycling** of metals and use of scrap metals and substitutes.
- Energy: use **public transport**, switch off electrical devices when not needed, use power-saving devices, and shift to non-conventional sources. The slogan: **‘Energy saved is energy produced’**.

Quick Revision: Likely Exam Questions

Likely question	Model answer
What is an ore?	Accumulation of a mineral mixed with other elements.
Which is the best iron ore?	Magnetite, with over 70 per cent iron.
Name the leading producer of manganese and bauxite.	Odisha.
Where is lignite found?	Neyveli in Tamil Nadu (also Rajasthan).
Name two nuclear power stations.	Tarapur and Kalpakkam.
What is a placer deposit?	Mineral accumulated in sands of valley floors and hills, e.g. gold.
Why should energy be conserved?	Fossil fuels are limited, and their use pollutes.

★ KEY TAKEAWAYS

- Minerals occur in **veins, beds, residual deposits, placers and sea water**.
- **Iron ore** (magnetite, haematite) is concentrated in Odisha-Jharkhand, Chhattisgarh and Karnataka; **Odisha** leads in manganese and bauxite.
- Coal ranks as **peat, lignite, bituminous and anthracite**; **Mumbai High** is the top oil field.
- Solar, wind, biogas, tidal and geothermal are non-conventional sources.
- **‘Energy saved is energy produced’**: conserve through efficiency, recycling and shift to renewables.